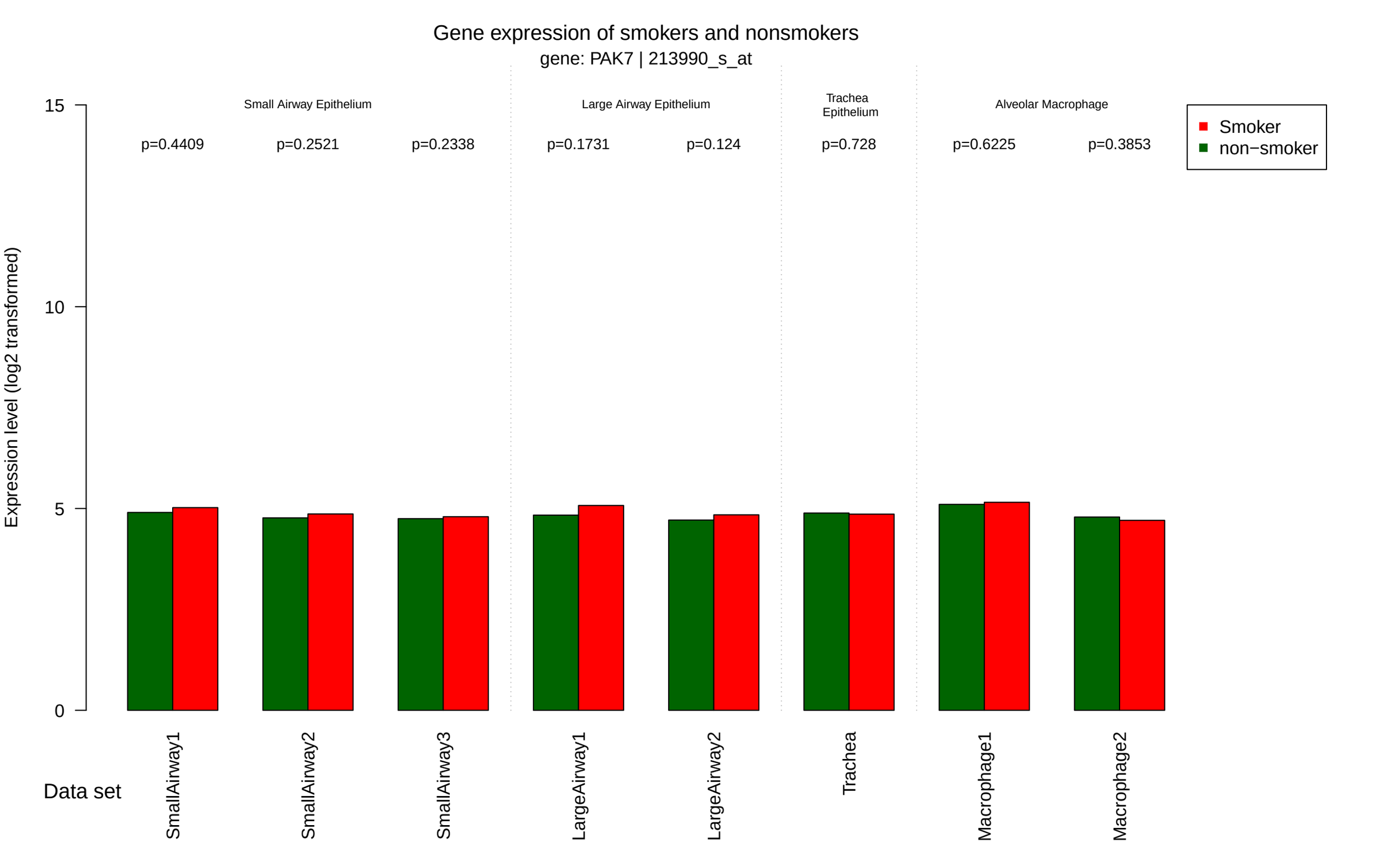
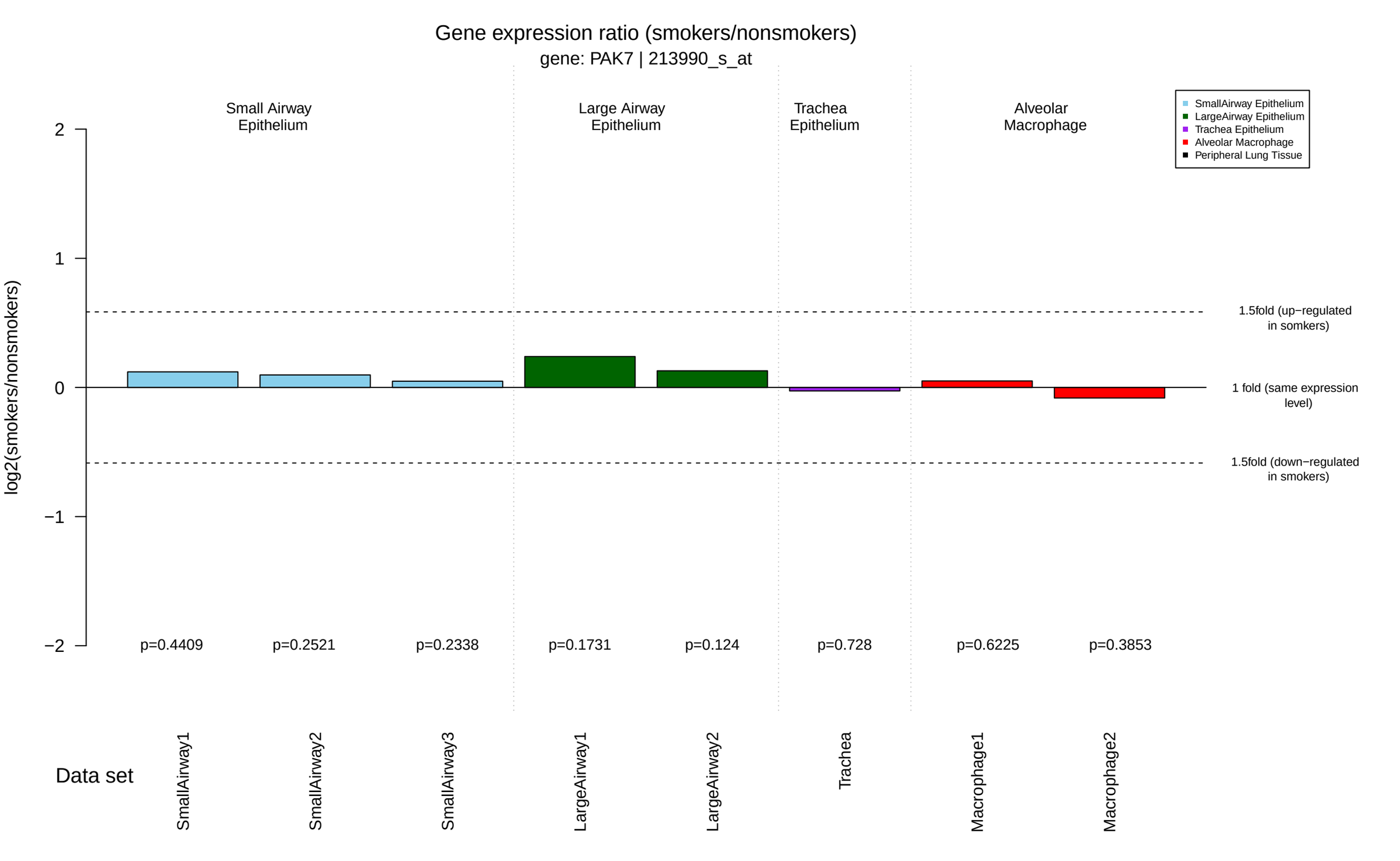
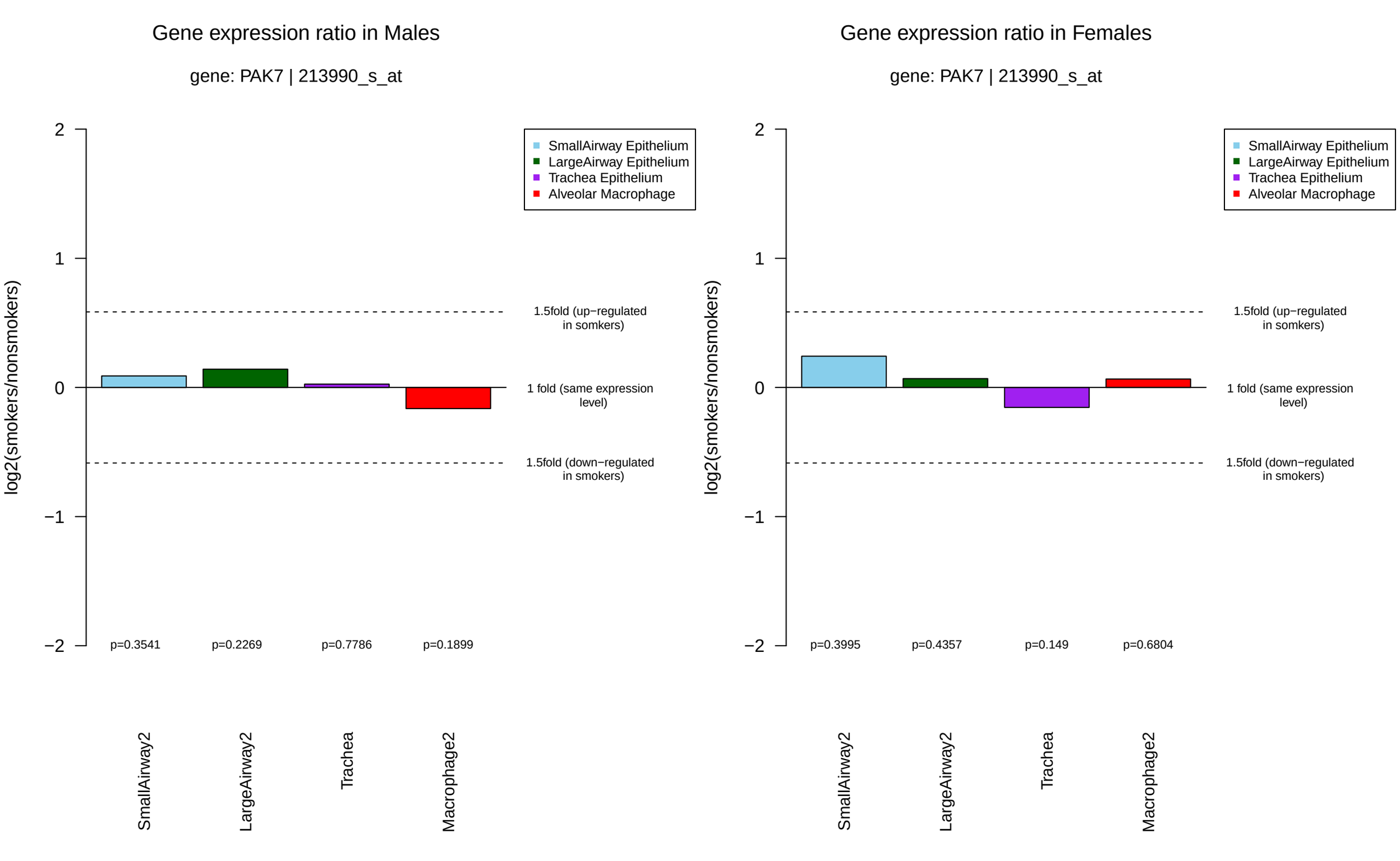
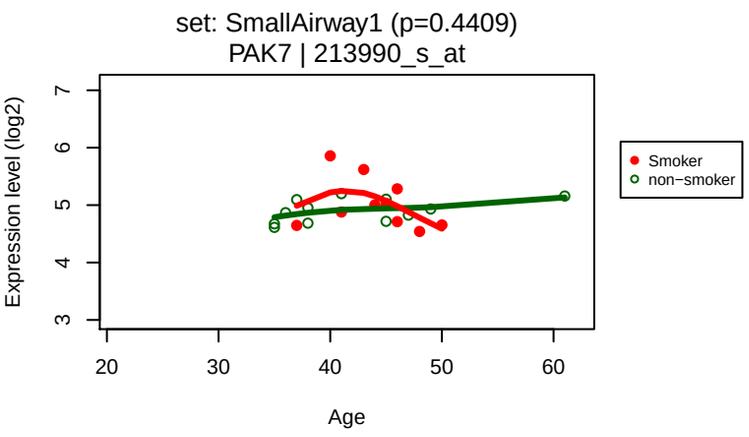
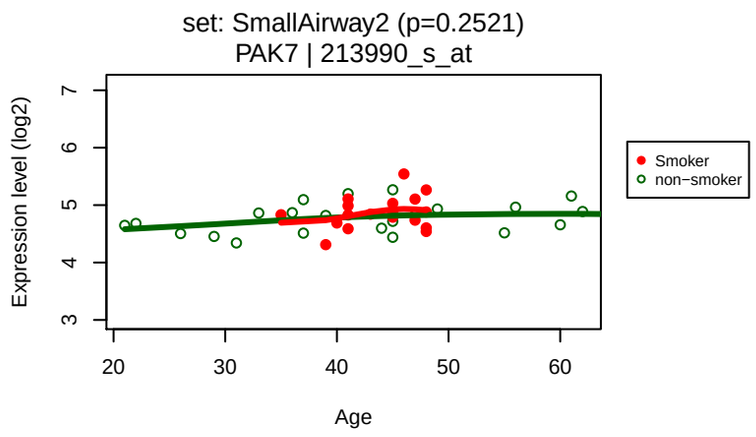
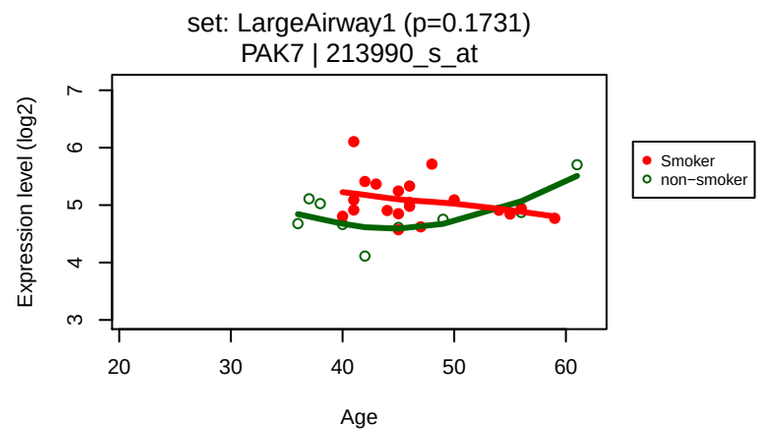

Gene expression of smokers and nonsmokers
gene: PAK7 | 213990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4409
p=0.2521
p=0.2338
p=0.1731
p=0.124
p=0.728
p=0.6225
p=0.3853
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

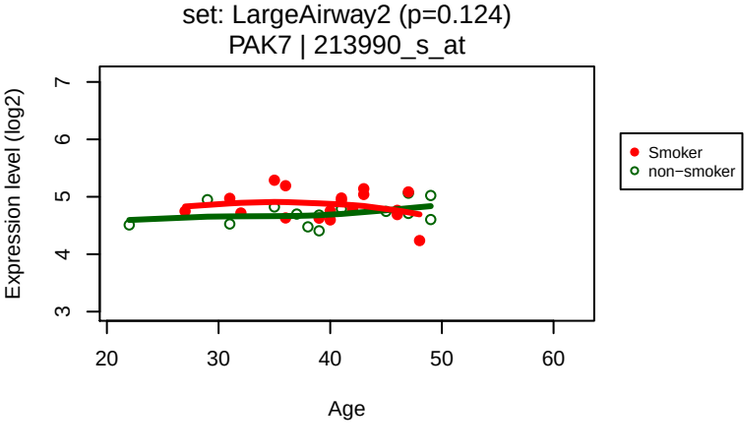

Gene expression ratio (smokers/nonsmokers)
gene: PAK7 | 213990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4409
p=0.2521
p=0.2338
p=0.1731
p=0.124
p=0.728
p=0.6225
p=0.3853
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

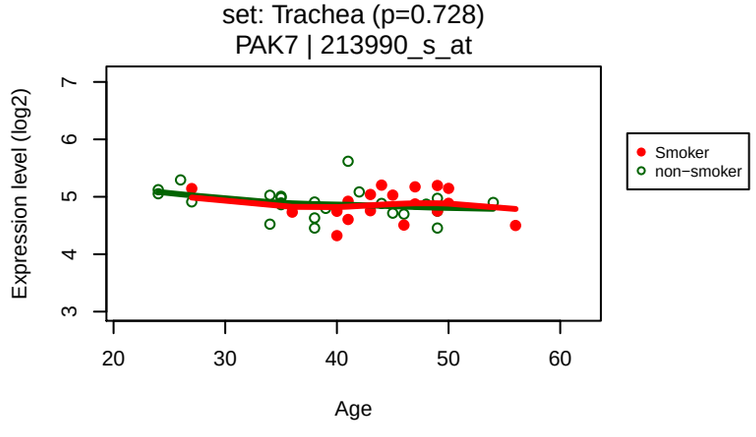

Gene expression ratio in Males
gene: PAK7 | 213990_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3541
p=0.2269
p=0.7786
p=0.1899
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK7 | 213990_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3995
p=0.4357
p=0.149
p=0.6804
SmallAirway2
LargeAirway2
Trachea
Macrophage2

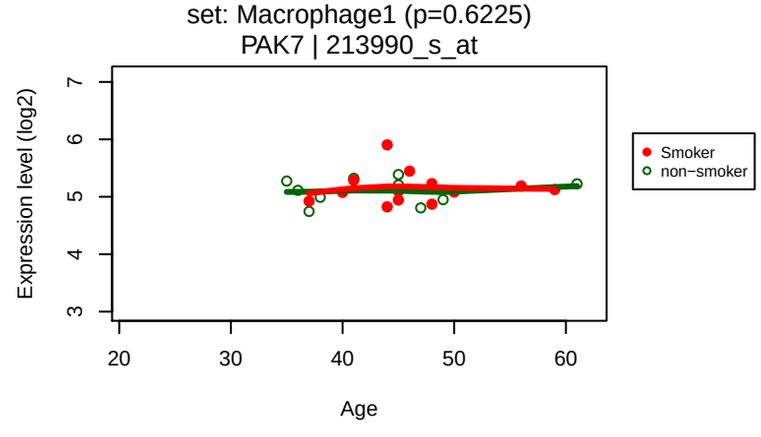

set: SmallAirway1 (p=0.4409)
PAK7 | 213990_s_at
Expression level (log2)
Age
Smoker
non−smoker

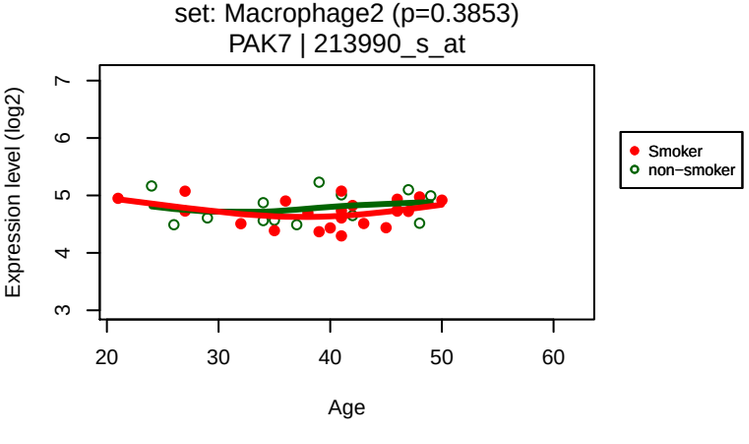

set: SmallAirway2 (p=0.2521)
PAK7 | 213990_s_at
Expression level (log2)
Age
Smoker
non−smoker

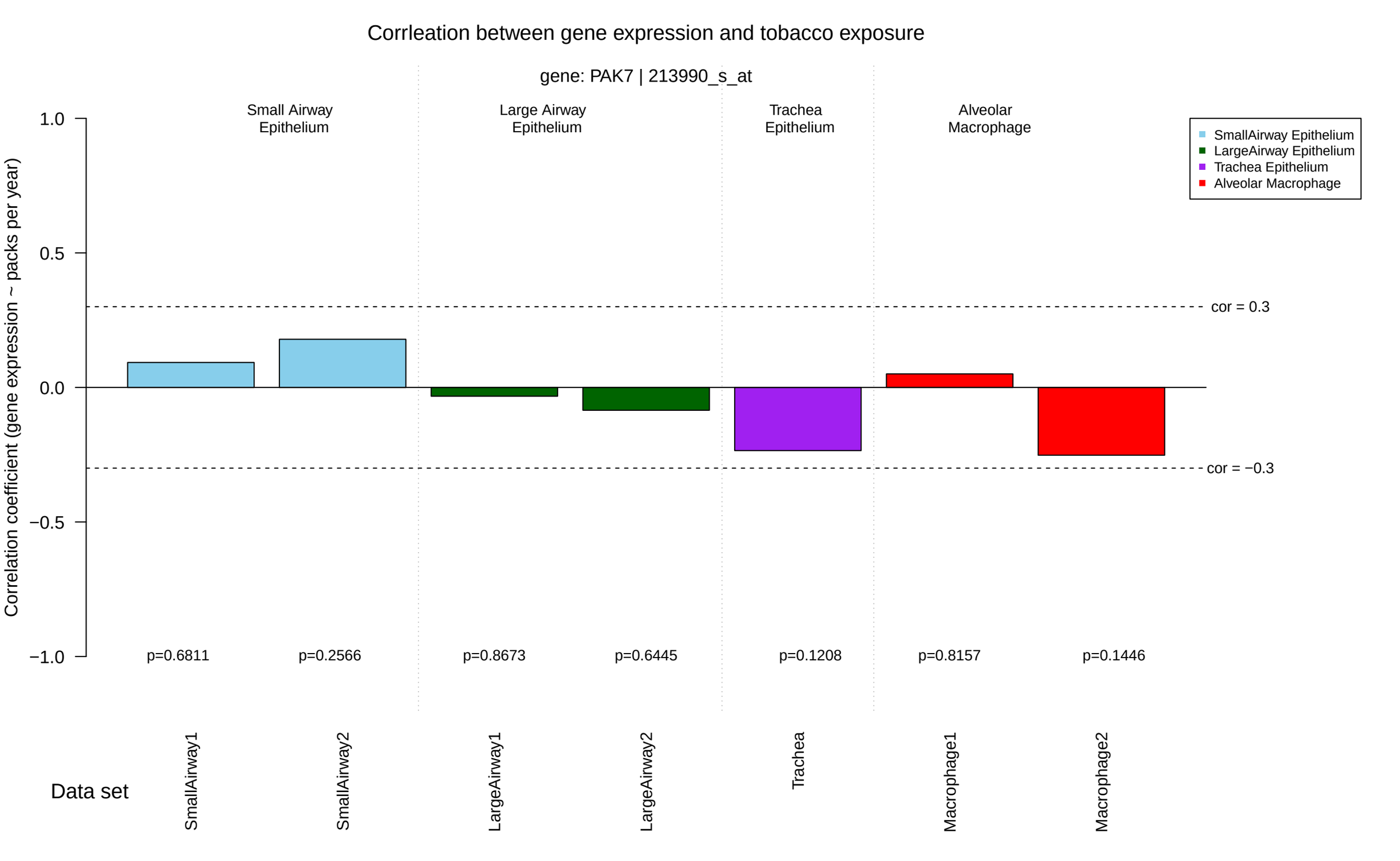

set: LargeAirway1 (p=0.1731)
PAK7 | 213990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.124)
PAK7 | 213990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.728)
PAK7 | 213990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6225)
PAK7 | 213990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3853)
PAK7 | 213990_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK7 | 213990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6811
p=0.2566
p=0.8673
p=0.6445
p=0.1208
p=0.8157
p=0.1446
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2